

CMPE 346 - Natural Language Processing

Assignment 02

Deadline: 24.05.2026 23.50

RULES

- This is an individual study. Each student is required to show the originality of their submission. Submissions involved in plagiarism will be penalized.
- Submit your implementations in a zip file named **[student_id_assignment02].zip**. Publish your trained model in Huggingface Hub and provide link in your README file. Make sure that your repository is **NOT** private and **publicly available**.
- Late submissions will **not** be accepted. The submission system will be closed after the deadline. Submissions by email will **not** be accepted.

Submission

- Your submission must follow the directory tree given below

```
.
├── data
│   ├── train.csv
│   └── valid.csv
├── main.py
├── model.py
├── preprocessing.py
├── run_test.py
└── README.md
```

Publish your trained model in Huggingface Hub and share the access link in README.md. Make sure that your model is publicly available.

ASSIGNMENT

Your task is to fine-tune a **Language Model** for multilingual binary sentiment classification. Training and validation sets and evaluation codes are shared in Learn. Details about the codes are shared in the files. Follow the instructions in the files; do not modify the evaluation codes. You can run the `run_test.py` script to evaluate your models. During the evaluation, a similar code will be executed. If your code fails in this script, you may not get points for the related tasks.

1. **(25 points)** Implement a preprocessor class for a multilingual Tokenizer approach.
2. **(25 points)** Implement a model class utilizing a Transformer Language Model for binary sentiment classification.
3. **(25 points)** Fine-tune a Transformer Language Model. Run different experiments to obtain the best model according to the validation set.
4. The remaining points for this assignment will be granted according to the performance of the models in the test set¹. Each model will be ranked relative to the submissions in the class. For example, if model's performance is ranked in the 3rd quartile, you will get 5 points additional to the points given above.

F1-Score in Quartile	Points
1st Quartile	25 points
2nd Quartile	15 points
3rd Quartile	5 points
4th Quartile	0 point

¹Baseline F1 score: 0.7958